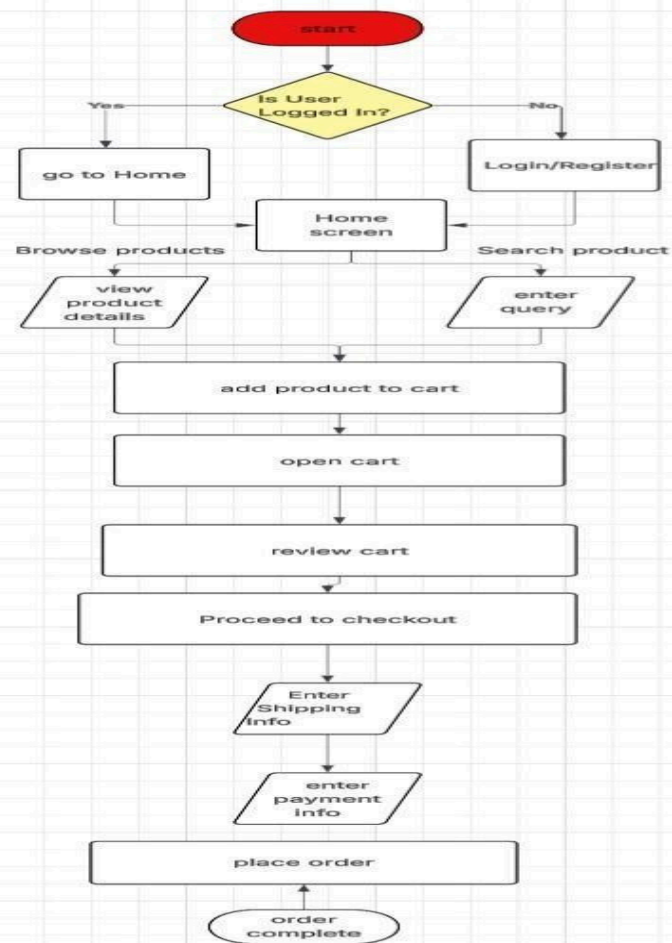


UID LAB EXPERIMENT – 4A

Flowchart Documentation for shopping app



1. Overview

This document explains the process flow represented in the diagram. The flow illustrates a typical **ecommerce user journey**, starting from login verification and proceeding through product browsing, cart management, checkout, and order completion.

2. Flow Summary

The process includes an initial decision (login check) followed by a linear shopping flow with optional browsing/search paths.

High-level sequence:

Start → Login Check → Home Screen → (Browse / Search) → Product Selection → Cart → Checkout → Shipping → Payment → Order Placement → End

3. Step-by-Step Explanation 3.1

Start

- Entry point of the system
- Represents the beginning of the user session

3.2 Decision Node: Is User Logged In? •

Checks whether the user is authenticated

Possible outcomes:

- Yes → Go to Home Screen
- No → Login/Register → Home Screen

3.3 Login / Register

- Allows new users to create an account
- Existing users can log in
- After successful authentication, user is redirected to Home Screen

3.4 Home Screen

- Central interface of the application
- Provides options to browse or search products
- Acts as the main navigation hub

3.5 Product Discovery (Two Paths)

A. Browse Products

- User navigates through product categories

- Leads to viewing product details

B. Search Product

- User enters a query
- System displays relevant products
- Both paths ultimately lead to product selection

3.6 View Product Details / Enter Query

- Displays detailed information about products ● Helps user make a purchase decision ● Includes price, description, etc.

3.7 Add Product to Cart

- Selected product is added to the shopping cart
- Acts as a temporary storage before purchase

3.8 Open Cart

- User accesses the cart
- Displays all selected items

3.9 Review Cart

- User verifies items, quantity, and price ● Can modify or remove items

3.10 Proceed to Checkout

- Initiates the purchase process
- Moves user toward final transaction steps

3.11 Enter Shipping Information

- User provides delivery details
- Includes address and contact information

3.12 Enter Payment Information

- User enters payment details
- Includes card, UPI, or other payment methods

3.13 Place Order:

- Final confirmation step
- Order is submitted to the system

3.14 Order Complete

- Indicates successful transaction
- Marks the end of the user journey

4. Flow Characteristics

- Initial **decision-based login validation** ● Dual paths for **product discovery (browse/search)**
- Linear progression after product selection
- Sequential checkout process
- Single endpoint (order completion)

5. Logical Interpretation

1. User enters the system and login status is checked
2. User is redirected to home screen after authentication
3. User searches or browses products
4. User selects and adds items to cart
5. Cart is reviewed before purchase
6. User enters shipping and payment details
7. Order is placed successfully
8. Process ends with order confirmation

6. Conclusion

The flowchart represents a **complete e-commerce purchasing workflow**, integrating authentication, product discovery, cart management, and transaction processing. The structure ensures a smooth and logical progression, making the system user-friendly and efficient.